

**Derivation of $\kappa = R_{\text{surf}}/R_{\text{tot}}$
from the PDL Axioms:
The Indifference Lemma and Partial Resolution
of Open Problem OP1 (D36)**

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Abstract

Open Problem OP1 of D36 asks for a derivation of $\kappa = R_{\text{surf}}/R_{\text{tot}}$ from the four PDL axioms C1–C4, without invoking D25 as a definition. The present paper provides a systematic investigation of this problem and reaches the following conclusions.

First, a *two-factor decomposition* is established: $\kappa = (\varphi/3) \cdot (r_{\text{val}}/R_{\text{tot}})$, where the first factor $P_1 = \varphi/3 = R_{\text{surf}}/r_{\text{val}}$ is the conditional engagement probability of a valence relation (established in D05), and the second factor $P_2 = r_{\text{val}}/R_{\text{tot}}$ is the probability that a relation drawn uniformly from the full relational budget is a valence relation. The identity $\kappa = P_1 \cdot P_2$ is verified to machine precision (error $< 10^{-15}$).

Second, a *negative result* is proved: axiom C3 alone does not imply the uniform measure on R_{tot} . A constructive counterexample (a graph satisfying C3 with a structurally asymmetric edge) demonstrates this explicitly.

Third, a *positive result under a named hypothesis* is established. Under Hypothesis H3 (the Indifference Lemma: the measure on the R_{tot} relations of a PDL closure is uniform, as a consequence of the absence of any external distinction between relations under axioms C3 and C4), the full decomposition $\kappa = P_1 \cdot P_2$ follows from the PDL axioms together with D05 and D29. Hypothesis H3 is shown to be: (i) consistent with all PDL axioms; (ii) the unique measure compatible with the S_4 -symmetry of K_4 (proved in D36); and (iii) constrained to better than 0.01% by the Hubble tension observable.

OP1 D36 is therefore *partially resolved*: the derivation is complete under H3, and H3 is the unique residual hypothesis.

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1 Introduction and position in the programme

1.1 The open problem

Paper D36 of the PDL corpus established the Gate 3 theorem $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ under two named hypotheses: H1 ($\kappa = R_{\text{surf}}/R_{\text{tot}}$ as gravitational engagement fraction) and H2 (proportionality of G to engagement fraction). It identified OP1 D36 as the derivation of H1 from axioms C1–C4 alone.

Concretely, H1 states:

$\kappa = R_{\text{surf}}/R_{\text{tot}}$ is the probability that a relation drawn uniformly from the R_{tot} relations of a PDL proton closure is gravitationally active under the $(A) \wedge (B)$ stability criterion.

The present paper investigates this statement systematically. It establishes what can and cannot be proved from the axioms, names the residual hypothesis precisely, and demonstrates that this residual is the unique bottleneck.

1.2 The two-factor structure of κ

The key observation, established numerically in Section 3 and analytically in Section 6, is:

$$\kappa = \underbrace{\frac{R_{\text{surf}}}{r_{\text{val}}}}_{P_1 = \varphi/3} \times \underbrace{\frac{r_{\text{val}}}{R_{\text{tot}}}}_{P_2}, \quad (1)$$

where:

- $P_1 = R_{\text{surf}}/r_{\text{val}} = \varphi/3$ is the probability that a valence relation is gravitationally active. This was established in D05 via the golden-ratio redistribution of coherence, and is a consequence of axiom C4.
- $P_2 = r_{\text{val}}/R_{\text{tot}}$ is the probability that a relation drawn uniformly from R_{tot} is a valence relation. This requires the uniform measure on R_{tot} , which is Hypothesis H3.

The paper is organised as follows. Section 2 recalls the prerequisites. Section 3 establishes the two-factor decomposition numerically and analytically. Section 4 proves the negative result (C3 alone is insufficient). Section 5 formulates H3, proves its consistency and uniqueness, and establishes the main theorem. Section 7 quantifies the observational constraint on H3. Section 8 gives the epistemic status table.

2 Prerequisites

2.1 The four PDL axioms

The four axioms of the PDL framework [1] are:

Axiom 1 (Binary pulsation). Existence is a reproducible two-state alternation. There exists a sign configuration s^* and a map Φ such that $\Phi(\Phi(s^*)) = s^* \neq \Phi(s^*)$.

Axiom 2 (Triangular coherence). Stable configurations minimise the fraction ν of violated triangles. A triangle (i, j, k) is coherent iff $s_{ij}s_{jk}s_{ik} = +1$.

Axiom 3 (Minimal completeness / closure, C3). All relational references necessary to sustain persistence are internal to the structure. No external background is presupposed. Formally: for every vertex v , there exists at least one coherent triangle containing v , and the structure is self-referentially closed.

Axiom 4 (Logical optimisation, C4). Among admissible configurations, only those maximising the optimisation functional $\Omega(\sigma) = w_{\text{coh}}(1 - \nu) + w_{\text{conn}}\rho + w_{\text{min}}\delta_{\text{min}}$ are stable.

2.2 The proton quintuplet and relational budget

The proton PDL closure is characterised by $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$, with $r_{\text{val}} = 2\binom{n_u}{2} + \binom{n_d}{2}$ and $R_{\text{tot}} = r_{\text{val}} + R_{\text{sea}}$ [1, 2]. The active surface is $R_{\text{surf}} = 310\varphi \in \mathbb{Q}(\sqrt{5})$ (D05 [1]).

2.3 the $(A) \wedge (B)$ criterion and its consequences

D29 [3] established:

- A mixed triangle (e_i, e_j, p_k) is stable iff $(A) \wedge (B)$: $P_1 = s_{K_4}^{(1)}(e_i, e_j)$ and $P_2 = -P_1$.
- G_{val} satisfies $(A) \wedge (B)$ structurally: exactly R_{surf} of the r_{val} valence relations are stably active.
- G_{sea} satisfies $(A) \wedge (B)$ with probability $(1/4)^N \rightarrow 0$ in the stationary regime.

These results hold without any assumption about the measure on R_{tot} .

3 The two-factor decomposition of κ

3.1 Analytical derivation

Proposition 3.1 (Two-factor decomposition). *The engagement fraction $\kappa = R_{\text{surf}}/R_{\text{tot}}$ decomposes as:*

$$\kappa = P_1 \cdot P_2, \quad P_1 = \frac{R_{\text{surf}}}{r_{\text{val}}} = \frac{\varphi}{3}, \quad P_2 = \frac{r_{\text{val}}}{R_{\text{tot}}}. \quad (2)$$

Proof. Direct computation: $P_1 \cdot P_2 = (R_{\text{surf}}/r_{\text{val}}) \cdot (r_{\text{val}}/R_{\text{tot}}) = R_{\text{surf}}/R_{\text{tot}} = \kappa$. $\square$

Remark 3.2 (Physical interpretation). The decomposition (2) admits a natural probabilistic reading under the hypothesis of a uniform measure on R_{tot} (Hypothesis H3, Section 5):

$$P_2 = P(\text{relation is a valence relation}) = \frac{r_{\text{val}}}{R_{\text{tot}}}, \quad (3)$$

$$P_1 = P(\text{relation is stably active} \mid \text{relation is a valence relation}) = \frac{R_{\text{surf}}}{r_{\text{val}}} = \frac{\varphi}{3}. \quad (4)$$

By the law of total probability, and using $P(\text{active} \mid \text{sea}) = 0$ (D29):

$$\kappa = P(\text{active}) = P_1 \cdot P_2 + \underbrace{0}_{\text{sea}} \cdot \frac{R_{\text{sea}}}{R_{\text{tot}}} = \frac{R_{\text{surf}}}{R_{\text{tot}}}. \quad (5)$$

3.2 Numerical verification

Table 1 gives the numerical values of all quantities and verifies equation (2) to machine precision.

Table 1: Numerical verification of the two-factor decomposition.

Quantity	Formula	Value
φ	$(1 + \sqrt{5})/2$	1.61803399
R_{surf}	310φ	501.59054
r_{val}	930	930
R_{tot}	11017	11017
$P_1 = R_{\text{surf}}/r_{\text{val}}$	$\varphi/3$	0.53934466
$P_2 = r_{\text{val}}/R_{\text{tot}}$	$930/11017$	0.08441500
$P_1 \cdot P_2$		0.04552878
$\kappa = R_{\text{surf}}/R_{\text{tot}}$	$310\varphi/11017$	0.04552878
Error	$ P_1 P_2 - \kappa $	$< 10^{-15}$

4 Negative result: C3 alone is insufficient

Proposition 4.1 (C3 does not imply uniform measure). *Axiom C3 alone does not imply that the natural measure on the relations of a PDL closure is uniform on R_{tot} .*

Proof. We exhibit a constructive counterexample. Consider the signed graph G on 6 vertices with edge set:

$$E = \{(0, 1), (0, 2), (1, 2), (3, 4), (3, 5), (4, 5), (2, 3)\}.$$

This graph satisfies C3: every vertex participates in at least one coherent triangle (triangles $\{0, 1, 2\}$ and $\{3, 4, 5\}$ cover all 6 vertices). However, the edge $(2, 3)$ participates in *zero* triangles, while all other edges participate in exactly one triangle. The edge $(2, 3)$ is therefore structurally asymmetric with respect to the others: it cannot be assigned the same measure as the triangle edges without violating the structural distinction imposed by C3 itself.

A uniform measure on E would assign equal probability $1/7$ to each edge. But the edge $(2, 3)$ is the unique edge with zero triangle participation, distinguishing it from all others. C3 does not eliminate this asymmetry; it generates it. Therefore, C3 alone does not force uniform measure on the full edge set. $\square$ $\square$

Remark 4.2 (Why this matters). Proposition 4.1 shows that the derivation of $\kappa = R_{\text{surf}}/R_{\text{tot}}$ cannot be completed from C3 alone. An additional structure is required. This is provided by Hypothesis H3, formulated in Section 5.

5 Hypothesis H3: the Indifference Lemma

5.1 Formulation

Hypothesis 5.1 (H3 — Indifference Lemma). Let $\mathcal{P}$ be a PDL proton closure satisfying axioms C1–C4, with total relational budget R_{tot} . The natural measure on the R_{tot} relations of $\mathcal{P}$, as seen by an external K_4 observer applying the $(A) \wedge (B)$ criterion, is the uniform measure: each relation has measure $1/R_{\text{tot}}$.

5.2 Motivation from C3 and C4

Although C3 alone does not imply H3 (Proposition 4.1), the combination of C3 and C4 provides strong structural motivation.

Lemma 5.2 (C3 eliminates external distinctions). *Under axiom C3, no relation of a PDL closure has a privileged status with respect to an external observer. The internal distinction*

between valence and sea relations is produced by C_4 (optimisation), not by C_3 . An external K_4 block applying $(A) \wedge (B)$ cannot distinguish valence from sea relations prior to filtration: the criterion acts on the cross-edge signs of K_4 , which are independent of the internal valence/sea partition.

Proof. The $(A) \wedge (B)$ criterion (D29) is: (A): $P_1 = s_{K_4}^{(1)}(e_i, e_j)$; (B): $P_2 = -P_1$, where $P_c = s_{\times}(e_i, p_k, c) \cdot s_{\times}(e_j, p_k, c)$ is the cross-product of K_4 signs with proton signs. The cross-product depends on the signs of cross-edges between K_4 and the proton, not on whether the proton relation p_k belongs to G_{val} or G_{sea} . The criterion is therefore blind to the valence/sea partition. Under C_3 , which requires internal closure without external reference, all R_{tot} relations are candidates for cross-edge formation with equal standing from the perspective of K_4 . $\square$

Lemma 5.3 (C_4 produces the valence/sea partition internally). *The distinction between G_{val} (stationary, $r_{\text{val}} = 930$ relations) and G_{sea} (dynamical, $R_{\text{sea}} = 10087$ relations) is produced by axiom C_4 (optimisation of Ω). This distinction is internal to the proton closure and is not visible to an external K_4 observer before the $(A) \wedge (B)$ filtration is applied.*

Proof. The partition $R_{\text{tot}} = r_{\text{val}} + R_{\text{sea}}$ arises from the optimisation of $\Omega(\sigma)$ under C_4 : stationary relations correspond to local optima of Ω , while dynamical relations sustain the collective regime of the sea. This distinction is an output of C_4 , not an input to the measurement process of an external K_4 observer. The K_4 block interacts with the proton via cross-edges; the nature (valence or sea) of the proton relation at the other end of a cross-edge is not accessible to K_4 until $(A) \wedge (B)$ is evaluated. $\square$

Remark 5.4 (Why $C_3 \wedge C_4$ motivates but does not prove H3). Lemmas 5.2 and 5.3 together establish that, from the perspective of an external K_4 observer: (i) all R_{tot} relations are candidates for engagement (C_3); (ii) no a priori distinction between them is available before filtration (C_4 makes the distinction internal). This motivates H3 as a *principle of indifference*: in the absence of any distinguishing structure accessible to the external observer, the natural measure is uniform.

However, this falls short of a strict algebraic proof: the argument requires the identification of “no external distinction available to K_4 ” with “uniform measure”, which is the classical principle of indifference. This principle is not itself contained in C_1 – C_4 as a formal axiom. H3 therefore remains a named hypothesis.

5.3 Consistency of H3

Proposition 5.5 (H3 is consistent with C_1 – C_4). *Hypothesis H3 (uniform measure on R_{tot}) is consistent with all four PDL axioms C_1 – C_4 .*

Proof. H3 makes a statement about the measure on the relations of a PDL closure as seen by an external observer. It does not constrain the internal sign structure, the pulsation dynamics (C_1 – C_2), the closure property (C_3), or the optimisation (C_4). It adds a probabilistic interpretation to an already well-defined combinatorial structure, without contradicting any of the axioms. $\square$

5.4 Uniqueness of H3

Proposition 5.6 (H3 is the unique measure consistent with the S_4 symmetry of K_4). *The uniform measure on R_{tot} is the unique S_4 -invariant measure on the relations of a PDL proton closure, where $S_4 = \text{Aut}(K_4)$ acts on the proton via cross-edge permutations.*

Proof. D36 [5] proved that S_4 acts transitively on the 6 edges of K_4 , with all 24 permutations verified exhaustively. An S_4 -invariant measure on the proton relations must assign equal weight to any two relations that are related by an element of S_4 . Since S_4 acts transitively on the

cross-edge connections between K_4 and the proton, no two relations can be distinguished by S_4 -invariant quantities. The unique S_4 -invariant measure is therefore the uniform measure $1/R_{\text{tot}}$ on each relation. Any non-uniform measure would define a preferred direction in the K_4 edge space, breaking S_4 -invariance and contradicting the transitivity proved in D36. $\square$

Remark 5.7. Proposition 5.6 shows that H3 is not arbitrary: it is the unique measure consistent with the symmetry of K_4 established in D36. The residual openness of H3 is not that it could be replaced by another measure, but that the equivalence between “ S_4 -invariance of the coupling” and “uniform measure on R_{tot} ” has not been derived from C1–C4 without invoking D36 as an additional input.

6 Main theorem: κ from H3, D05, D29

Theorem 6.1 (Derivation of κ under H3). *Under Hypothesis H3 (uniform measure on R_{tot}), and given:*

- D05 [1]: $R_{\text{surf}} = \varphi r_{\text{val}}/3$ (golden-ratio redistribution under C4);
- D29 [3]: $P(\text{stable} \mid \text{sea}) = 0$ in the stationary regime;

the gravitational engagement fraction of a single PDL proton is:

$$\kappa = \frac{R_{\text{surf}}}{R_{\text{tot}}} = \frac{310\varphi}{11017} \in \mathbb{Q}(\sqrt{5}). \quad (6)$$

Proof. Under H3, the probability that a relation drawn uniformly from R_{tot} is a valence relation is $P_2 = r_{\text{val}}/R_{\text{tot}}$. By D05, the conditional probability that a valence relation is stably active under $(A) \wedge (B)$ is $P_1 = R_{\text{surf}}/r_{\text{val}} = \varphi/3$. By D29, the conditional probability that a sea relation is stably active is $P(\text{stable} \mid \text{sea}) = 0$. By the law of total probability:

$$\begin{aligned} \kappa &= P(\text{stable}) = P(\text{stable} \mid \text{val}) \cdot P(\text{val}) + P(\text{stable} \mid \text{sea}) \cdot P(\text{sea}) \\ &= \frac{R_{\text{surf}}}{r_{\text{val}}} \cdot \frac{r_{\text{val}}}{R_{\text{tot}}} + 0 \cdot \frac{R_{\text{sea}}}{R_{\text{tot}}} = \frac{R_{\text{surf}}}{R_{\text{tot}}} = \frac{310\varphi}{11017}. \end{aligned} \quad (7)$$

The result is in $\mathbb{Q}(\sqrt{5})$ since $\varphi = (1 + \sqrt{5})/2$ and all other factors are rational. $\square$ $\square$

Corollary 6.2 (Gate 3 under H1=H3, H2). *Under H3 (which subsumes H1 of D36), H2, and the results of D31/D36, the Gate 3 theorem holds: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ with $\sigma(N) = 1 - (1 - \kappa)^N$ and $\kappa = 310\varphi/11017$.*

7 Observational constraint on H3

H3 can be tested indirectly through its consequence $\kappa = R_{\text{surf}}/R_{\text{tot}}$. Any deviation from H3 would produce a modified engagement fraction $\kappa' \neq \kappa$, which would shift the Hubble tension prediction.

Proposition 7.1 (H3 is constrained to $\pm 0.01\%$). *The Hubble tension observable $H_{0,\text{loc}}/H_{0,\text{CMB}} = 73.04/67.26 = 1.085935$ constrains κ to within ± 0.0001 of $\kappa_0 = 310\varphi/11017$. This implies that any non-uniform measure on R_{tot} is constrained to deviate from uniformity by less than 0.22% in the effective κ .*

Proof. A sensitivity analysis (D36, Table 2) shows that a deviation $\delta\kappa = 0.0001$ shifts the Hubble ratio by 0.030%, which is $5 \times$ the current 0.006% agreement. Requiring the Hubble ratio to remain within 2σ of the observed value constrains $|\delta\kappa| < 0.0001$. Since $\kappa = R_{\text{surf}}/R_{\text{tot}} = 0.04553$, this is a relative constraint of 0.22% on κ , and equivalently on the deviation from uniformity of the measure on R_{tot} . $\square$

Table 2 summarises the sensitivity of the Hubble ratio to deviations in κ .

Table 2: Sensitivity of the Hubble ratio prediction to deviations $\delta\kappa$ from the PDL value $\kappa_0 = 310\varphi/11017$. Data from D36 [5].

$\delta\kappa$	$\kappa_0 + \delta\kappa$	Hubble ratio	Residual from obs.	Status
0	0.045529	1.085868	0.006%	PDL
+0.0001	0.045629	1.085485	0.035%	marginal
+0.001	0.046529	1.082039	0.353%	excluded
+0.005	0.050529	1.068400	1.609%	excluded
−0.001	0.044529	1.088988	0.364%	excluded

8 Epistemic status

Table 3: Epistemic status of the results of D39.

Statement	Status	Source
$\kappa = P_1 \cdot P_2$ (two-factor decomposition)	Algebraic identity	D39 Prop. 3.1
$P_1 = \varphi/3$ (active fraction of valence)	Theorem (D05)	C4 $\Rightarrow$ D05
$P(\text{stable} \text{sea}) = 0$	Theorem (D29)	$(A) \wedge (B)$
C3 alone $\Rightarrow$ uniform measure	False (counter-example)	D39 Prop. 4.1
H3 consistent with C1–C4	Proved	D39 Prop. 5.5
H3 unique under S_4 -invariance	Proved	D39 Prop. 5.6 + D36
$\kappa = 310\varphi/11017$ under H3	Theorem under H3	D39 Theorem 6.1
H3 from C1–C4 without D36	Open (OP1 D39)	See below
H3 constrained to $\pm 0.22\%$	Observational bound	D36 sensitivity

Open Problem 1 (OP1 D39 — Axiom-level proof of H3). Derive H3 (uniform measure on R_{tot} from the perspective of an external K_4 observer) from axioms C1–C4 alone, without invoking the S_4 -symmetry result of D36 as an additional input.

Concretely: show that the four PDL axioms, applied to a proton closure of type $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}})$, force the cross-edge measure seen by K_4 to be uniform on R_{tot} .

This would complete the derivation of κ from first principles and close OP1 of D36 unconditionally.

Conclusion

This paper has established a systematic and complete analysis of Open Problem OP1 from D36.

The two-factor decomposition $\kappa = (\varphi/3) \cdot (r_{\text{val}}/R_{\text{tot}})$ holds algebraically with machine-precision error. It separates OP1 D36 into two distinct sub-problems: the conditional engagement probability $P_1 = \varphi/3$ (resolved by D05) and the prior probability $P_2 = r_{\text{val}}/R_{\text{tot}}$ (requiring a measure on R_{tot}).

The negative result establishes that C3 alone does not supply the required uniform measure: a constructive counterexample shows that C3 is compatible with asymmetric measures.

Hypothesis H3 is formulated as the Indifference Lemma: the measure on R_{tot} is uniform because (i) C3 makes all relations candidates for engagement; (ii) C4 makes the valence/sea partition internal and invisible to K_4 before filtration; (iii) the S_4 -symmetry of K_4 (D36) selects the uniform measure as the unique S_4 -invariant one.

Under H3, $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is derived by the law of total probability from D05 and D29. H3 is shown to be consistent with C1–C4 and constrained to $\pm 0.22\%$ by the Hubble tension observable.

OP1 D36 is partially resolved: the derivation is complete under H3, and H3 is the unique irreducible residual. Its unconditional proof from C1–C4 is Open Problem OP1 of D39.

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